SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I M.Himanath sai (192525246) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:M.Himanath sai

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Question 1

Given:

- File size = **150 MB**
- Bandwidth = **50 Mbps**
- Segment size = **1500 bytes**
- Data Link overhead = **40 bytes/frame**

Step 1: Convert File Size into Bytes

150 MB = 150 × 1024 × 1024

= **157,286,400 bytes**

Step 2: Calculate Number of Segments

Number of Segments = Total File Size ÷ Segment Size

= 157,286,400 ÷ 1500

= **104,858 segments** (approx.)

Since one segment forms one frame,

Number of Frames = 104,858

Step 3: Calculate Total Data Link Layer Overhead

Overhead per frame = **40 bytes**

Total Overhead

= 104,858 × 40

= **4,194,320 bytes**

≈ **4 MB**

Step 4: Calculate Total Data Transmitted

Total Data

= Original Data + Overhead

= 157,286,400 + 4,194,320

= **161,480,720 bytes**

Convert to bits

= 161,480,720 × 8

= **1,291,845,760 bits**

Step 5: Calculate Transmission Time

Bandwidth = 50 Mbps

= 50 × 10^6 bits/sec

Transmission Time

= Total Bits ÷ Bandwidth

= 1,291,845,760 ÷ 50,000,000

= **25.84 seconds**

Final Answers

- **Total Segments = 104,858**
- **Total Frames = 104,858**
- **Total Overhead = 4,194,320 bytes (≈4 MB)**
- **Total Transmission Time = 25.84 seconds**

Explanation

The **Transport Layer** divides the large file into smaller segments, assigns sequence numbers, performs error detection, acknowledgments, and retransmissions to ensure reliable delivery. The **Data Link Layer** encapsulates each segment into frames, adds headers and trailers for addressing and error checking (using CRC), and detects transmission errors. Together, these layers provide reliable, accurate, and efficient communication across the network.

Question 2

Given:

- Sliding Window Size = **6 frames**
- Data per Frame = **1024 bytes**
- Total Frames = **48 frames**

Step 1: Calculate Number of Transmission Windows

Formula:

$$\text{Number of Windows} = \frac{\text{Total Frames}}{\text{Window Size}}$$

Substitute the values:

$$= \frac{48}{6} = 8$$

Number of Transmission Windows = 8

Step 2: Calculate Total Data Transmitted

Formula:

$$\text{Total Data} = \text{Total Frames} \times \text{Frame Size}$$

$$= 48 \times 1024$$

$= 49{,}152 \text{ bytes}$

Total User Data = 49,152 bytes

Step 3: Calculate Retransmissions

One frame is lost in every transmission window.

Number of Windows = **8**

Therefore,

Retransmissions = **8 frames**

Step 4: Calculate Retransmitted Data

8×1024

$= 8192 \text{ bytes}$

Retransmitted Data = 8,192 bytes

Final Answers

- **Sliding Window Size = 6 frames**
- **Total Frames = 48**
- **Number of Transmission Windows = 8**
- **Total User Data = 49,152 bytes**
- **Frames Retransmitted = 8**
- **Retransmitted Data = 8,192 bytes**

Explanation

Flow control is a Transport Layer mechanism that regulates the amount of data a sender can transmit before receiving an acknowledgment. Using the sliding window protocol, the sender can transmit multiple frames continuously without waiting for an acknowledgment after each frame. This improves channel utilization and increases throughput. If a frame is lost, only the lost frame is retransmitted instead of all frames, reducing unnecessary retransmissions and improving network efficiency. Flow control also prevents the receiver's buffer from overflowing, ensuring reliable and efficient communication.

Question 3

Given:

- IP Packet Size = **12,000 bytes**
- MTU = **1500 bytes**
- IP Header per Fragment = **20 bytes**

Step 1: Calculate Maximum Data per Fragment

Each fragment must include a 20-byte IP header.

Data per Fragment

= MTU − Header

= 1500 − 20

= **1480 bytes**

Step 2: Calculate Number of Fragments

Formula:

$$\text{Number of Fragments} = \left\lceil \frac{\text{Packet Size}}{\text{Data per Fragment}} \right\rceil$$

Substitute the values:

$$= \frac{12000}{1480} = 8.11$$

Since a fraction of a fragment is not possible,

Number of Fragments = 9

Step 3: Calculate Total Header Overhead

Formula:

$$\text{Header Overhead} = \text{Number of Fragments} \times \text{Header Size}$$

$$= 9 \times 20$$

$$= 180 \text{ bytes}$$

Step 4: Calculate Data in the Last Fragment

Data carried by first 8 fragments

= 8 × 1480

= **11,840 bytes**

Remaining data

= 12,000 − 11,840

= **160 bytes**

Last fragment size

= 160 + 20

= **180 bytes**

Final Answers

- **Maximum Data per Fragment = 1480 bytes**
- **Number of Fragments = 9**
- **Header Overhead = 180 bytes**
- **Last Fragment Size = 180 bytes**

Explanation

The **Network Layer** is responsible for routing packets between the source and destination. When a packet is larger than the MTU of a network, it performs **fragmentation**, dividing the packet into smaller fragments. Each fragment receives its own IP header containing identification, fragment offset, and flags. These fields allow the destination host to correctly reassemble all fragments into the original packet.

Question 4

Given

- Window size = 6 frames
- Total frames = 48
- Data per frame = 1024 bytes
- One frame lost in every window

Formula

Number of Windows = Total Frames ÷ Window Size

Calculation

Number of Windows

= 48 ÷ 6

= **8 windows**

One frame is lost in every window.

Retransmissions

= **8 frames**

Total User Data Sent

= 48 × 1024

= **49,152 bytes**

Explanation

The Sliding Window Protocol allows multiple frames to be transmitted before acknowledgements are received. This improves bandwidth utilization and reduces waiting time.

Flow control ensures that the sender does not send data faster than the receiver can process. It prevents receiver buffer overflow, minimizes congestion, and increases network efficiency.

If a frame is lost, only the missing frame is retransmitted, ensuring reliable communication.

Final Answer

- Number of transmission windows = **8**
- Total retransmissions = **8 frames**
- Total user data transmitted = **49,152 bytes**

Conclusion

Flow control improves communication by regulating transmission speed and preventing data loss while maximizing network performance.

Question 5

Given

- File size = 600 MB
- Checkpoint every 60 MB
- Failure after 390 MB

Formula

Number of Checkpoints

= Data Transferred ÷ Checkpoint Interval

Calculation

390 ÷ 60

= **6 checkpoints**

The checkpoints occur at:

60 MB

120 MB

180 MB

240 MB

300 MB

360 MB

Last checkpoint = **360 MB**

Data retransmitted

= 390 − 360

= **30 MB**

Explanation

The Session Layer establishes, maintains, and terminates communication sessions.

Synchronization points (checkpoints) allow interrupted communication to restart from the most recent checkpoint instead of retransmitting the complete file.

This saves bandwidth, reduces transmission time, and improves network reliability.

Final Answer

- Number of checkpoints = **6**
- Data retransmitted = **30 MB**

Conclusion

Synchronization is an important Session Layer feature because it enables efficient recovery after failures and minimizes retransmission.

Question 6

Given

- Original file = 900 MB
- Compression = 40%
- Encryption overhead = 5%

Calculation

Compressed Size

= 900 × (100 − 40)%

= 900 × 0.60

= **540 MB**

Encryption Overhead

= 540 × 5%

= **27 MB**

Encrypted File Size

= 540 + 27

= **567 MB**

Total Reduction

= 900 − 567

= **333 MB**

Percentage Reduction

= (333 ÷ 900) × 100

= **37%**

Explanation

The Presentation Layer is responsible for:

- Data compression
- Data encryption
- Data formatting
- Data translation

Compression reduces the file size, decreasing transmission time and bandwidth usage.

Encryption protects the confidentiality of data by converting it into unreadable ciphertext, ensuring secure communication over the network.

Final Answer

- Compressed file size = **540 MB**
- Encrypted file size = **567 MB**
- Total reduction = **37%**

Conclusion

The Presentation Layer improves network efficiency by compressing data and enhances security through encryption, ensuring faster and secure data transmission.

Question 7

Given

- Total data = **3 GB = 3000 MB**
- Request size = **3 MB**
- Throughput = **120 Mbps**

Formula

Number of Requests = Total Data ÷ Request Size

Transmission Time = (Data × 8) ÷ Bandwidth

Calculation

Number of Requests

= 3000 ÷ 3

= **1000 requests**

Convert data into megabits

= 3000 × 8

= **24,000 Mb**

Transmission Time

= 24,000 ÷ 120

= **200 seconds**

Explanation

The **Application Layer** is the top layer of the OSI model and provides network services directly to end users.

Its functions include:

- File Transfer (FTP)
- Email (SMTP)
- Web Browsing (HTTP)
- Domain Name Service (DNS)
- Remote Login (Telnet)

It provides reliable communication through authentication, file management, error reporting, and user-friendly network services.

Final Answer

- Number of requests = **1000**
- Transmission time = **200 seconds**

Conclusion

The Application Layer enables users to access network services efficiently while ensuring reliable file transfer through protocols such as FTP.

Question 8

Given

- Application delay = **4 ms**
- Transport delay = **3 ms**
- Network delay = **5 ms**
- Data Link delay = **2 ms**
- Physical delay = **1 ms**
- Propagation delay = **18 ms**
- Transmission delay = **12 ms**

Calculation

Processing Delay

= 4 + 3 + 5 + 2 + 1

= **15 ms**

Total Delay

= Processing + Propagation + Transmission

= 15 + 18 + 12

= **45 ms**

Explanation

Each OSI layer contributes as follows:

- **Application Layer:** Provides services such as FTP, HTTP, and Email.
- **Transport Layer:** Performs segmentation, flow control, and error recovery.
- **Network Layer:** Determines routing and logical addressing.

- **Data Link Layer:** Performs framing and error detection.
- **Physical Layer:** Transmits bits through cables or wireless media.

Propagation delay is the time taken for the signal to travel through the transmission medium, while transmission delay is the time required to place all bits onto the communication channel.

Final Answer

- Processing delay = **15 ms**
- Total end-to-end delay = **45 ms**

Conclusion

The total communication delay is the sum of processing, transmission, and propagation delays. Every OSI layer contributes to reliable and efficient communication.

Question 9

Given

- Useful data = **80 MB = 80,000,000 bytes**
- Frame size = **1600 bytes**
- Overhead = **80 bytes**

Calculation

Useful Payload per Frame

= 1600 − 80

= **1520 bytes**

Number of Frames

= 80,000,000 ÷ 1520

≈ **52,632 frames**

Total Overhead

= 52,632 × 80

= **4,210,560 bytes**

≈ **4.21 MB**

Transmission Efficiency

= (1520 ÷ 1600) × 100

= **95%**

Explanation

Protocol overhead includes headers and trailers added by different OSI layers for addressing, routing, sequencing, and error detection.

Although overhead consumes bandwidth, it is necessary for:

- Reliable communication

- Error detection
- Packet routing
- Flow control
- Data integrity

Higher overhead reduces transmission efficiency, while lower overhead improves bandwidth utilization.

Final Answer

- Frames required = **52,632**
- Total overhead = **4.21 MB**
- Transmission efficiency = **95%**

Conclusion

Protocol overhead slightly reduces efficiency but is essential for reliable and secure communication across networks.

Question 10

Given

- Original file = **240 MB**
- Compression = **30%**
- Segment size = **1400 bytes**
- Overhead per frame = **60 bytes**

Calculation

Compressed File Size

= 240 × 70%

= **168 MB**

Compressed Data

= **168,000,000 bytes**

Number of Segments

= 168,000,000 ÷ 1400

= **120,000 segments**

Protocol Overhead

= 120,000 × 60

= **7,200,000 bytes**

= **7.2 MB**

Final Data Transmitted

= 168 + 7.2

= **175.2 MB**

Explanation

The OSI layers work together as follows:

- **Presentation Layer:** Compresses data and performs encryption to improve efficiency and security.
- **Transport Layer:** Divides data into segments, performs sequencing, flow control, and error recovery.
- **Network Layer:** Determines the best route using logical IP addressing.
- **Data Link Layer:** Creates frames, adds headers/trailers, and performs error detection using CRC.
- **Physical Layer:** Transmits the final bit stream over wired or wireless communication media.

This layered architecture ensures secure, reliable, and efficient communication in IoT healthcare applications.

Final Answer

- Compressed file size = **168 MB**
- Number of segments = **120,000**
- Protocol overhead = **7.2 MB**
- Final transmitted data = **175.2 MB**

Conclusion

The OSI model enables efficient communication by assigning specific responsibilities to each layer. Compression reduces bandwidth usage, segmentation ensures reliable delivery, framing provides error detection, and the Physical Layer transmits the data successfully.